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main.c

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```c
#include <stdio.h>

void merge(int arr[], int left, int mid, int right) {
    int n1 = mid - left + 1;
    int n2 = right - mid;

    int L[100000], R[100000];

    for (int i = 0; i < n1; i++) L[i] = arr[left + i];
    for (int j = 0; j < n2; j++) R[j] = arr[mid + 1 + j];

    int i = 0, j = 0, k = left;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) arr[k++] = L[i++];
        else arr[k++] = R[j++];
    }
    while (i < n1) arr[k++] = L[i++];
    while (j < n2) arr[k++] = R[j++];
}
void mergeSort(int arr[], int left, int right) {
    if (left < right) {
        int mid = (left + right) / 2;

        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);
        merge(arr, left, mid, right);
    }
}
int main() {
    int n;
    printf("Enter number of students: ");
    scanf("%d", &n);
    int marks[100000];
    printf("Enter %d marks: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &marks[i]);
    }
    mergeSort(marks, 0, n - 1);
    printf("Sorted marks: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", marks[i]);
    }
    printf("\n");
    return 0;
}
```

Output

Clear

```
Enter number of students: 6
Enter 6 marks: 33 65 43 23 56 98
Sorted marks: 23 33 43 56 65 98


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```

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    int n1 = mid - left + 1;
    int n2 = right - mid;
    int L[100000], R[100000];
    for (int i = 0; i < n1; i++) L[i] = arr[left + i];
    for (int j = 0; j < n2; j++) R[j] = arr[mid + 1 + j];
    int i = 0, j = 0, k = left;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) arr[k++] = L[i++];
        else arr[k++] = R[j++];
    }
    while (i < n1) arr[k++] = L[i++];
    while (j < n2) arr[k++] = R[j++];
}
void mergeSort(int arr[], int left, int right) {
    if (left < right) {
        int mid = (left + right) / 2;
        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);
        merge(arr, left, mid, right);
    }
}
int main() {
    int n;
    printf("Enter number of products: ");
    scanf("%d", &n);
    int price[800000];
    printf("Enter %d prices: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &price[i]);
    }
    mergeSort(price, 0, n - 1);
    printf("Sorted prices: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", price[i]);
    }
    printf("\n");
    return 0;
}
```

Output

Clear

```
Enter number of products: 5
Enter 5 prices: 23 55 76 99 11
Sorted prices: 11 23 55 76 99


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```

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#include <stdio.h>

void merge(int arr[], int left, int mid, int right) {
    int n1 = mid - left + 1;
    int n2 = right - mid;
    int L[100000], R[100000];
    for (int i = 0; i < n1; i++) L[i] = arr[left + i];
    for (int j = 0; j < n2; j++) R[j] = arr[mid + 1 + j];
    int i = 0, j = 0, k = left;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) arr[k++] = L[i++];
        else arr[k++] = R[j++];
    }
    while (i < n1) arr[k++] = L[i++];
    while (j < n2) arr[k++] = R[j++];
}
void mergeSort(int arr[], int left, int right) {
    if (left < right) {
        int mid = (left + right) / 2;
        mergeSort(arr, left, mid);
        mergeSort(arr, mid + 1, right);
        merge(arr, left, mid, right);
    }
}
int main() {
    int n;
    printf("Enter number of employees: ");
    scanf("%d", &n);
    int salary[100000];
    printf("Enter %d salaries: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &salary[i]);
    }
    mergeSort(salary, 0, n - 1);
    printf("Sorted salaries: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", salary[i]);
    }
    printf("\n");
    int lowest = salary[0];
    int highest = salary[n - 1];
    int difference = highest - lowest;
    printf("Lowest Salary: %d\n", lowest);
    printf("Highest Salary: %d\n", highest);
    printf("Difference: %d\n", difference);
    return 0;
}
```

Output

Clear

```
Enter number of employees: 9
Enter 9 salaries: 764 46765534 57467 3456346 83456 247 873446 7974 134678
Sorted salaries: 247 764 7974 57467 83456 134678 873446 3456346 46765534
Lowest Salary: 247
Highest Salary: 46765534
Difference: 46765287


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```

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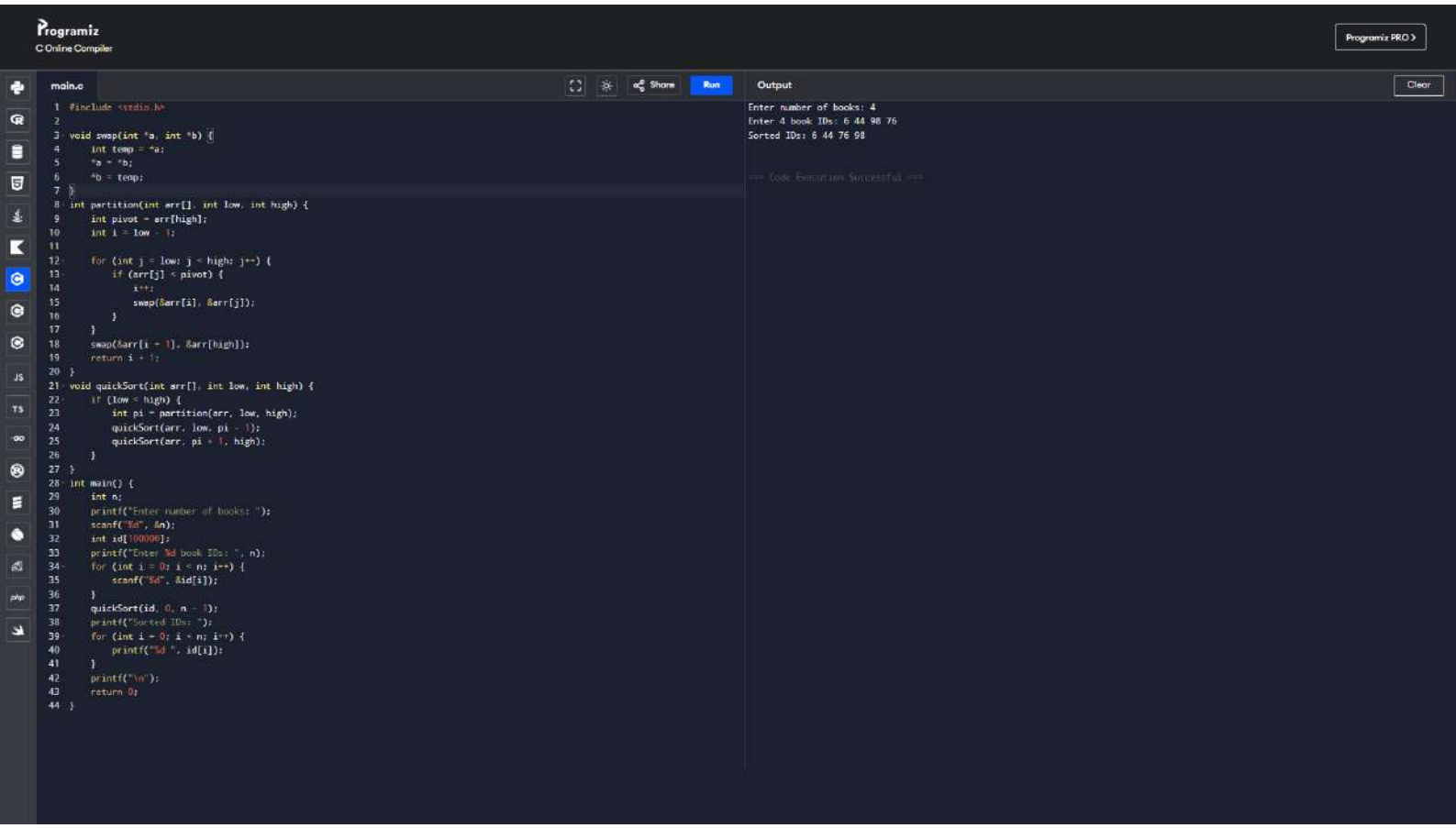
```c
#include <stdio.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = low - 1;
    for (int j = low; j < high; j++) {
        if (arr[j] < pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return i + 1;
}
void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}
int main() {
    int n;
    printf("Enter number of runners: ");
    scanf("%d", &n);
    int time[100000];
    printf("Enter %d timings: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &time[i]);
    }
    quickSort(time, 0, n - 1);
    printf("Sorted timings: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", time[i]);
    }
    printf("\n");
    return 0;
}
```

Output

Clear

```
Enter number of runners: 6
Enter 6 timings: 6 5 2 9 8 4
Sorted tinings: 2 4 5 6 8 9


=== Code Execution Successful ===
```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#include <stdio.h>
void swap(int *a, int *b) {
int temp = *a;
*a = *b;
*b = temp;
}
int partition(int arr[], int low, int high) {
int pivot = arr[high];
int i = low - 1;
for (int j = low; j < high; j++) {
if (arr[j] < pivot) {
i++;
swap(&arr[i], &arr[j]);
}
}
swap(&arr[i + 1], &arr[high]);
return i + 1;
}
void quickSort(int arr[], int low, int high) {
if (low < high) {
int pi = partition(arr, low, high);
quickSort(arr, low, pi - 1);
quickSort(arr, pi + 1, high);
}
}
int main() {
int n;
printf("Enter number of books: ");
scanf("%d", &n);
int id[100000];
printf("Enter %d book IDs: ", n);
for (int i = 0; i < n; i++) {
scanf("%d", &id[i]);
}
quickSort(id, 0, n - 1);
printf("Sorted IDs: ");
for (int i = 0; i < n; i++) {
printf("%d ", id[i]);
}
printf("\n");
return 0;
}
Enter number of books: 4
Enter 4 book IDs: 6 44 98 76
Sorted IDs: 6 44 76 98
=== Code Execution Successful ===

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```c
#include <stdio.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = low - 1;
    for (int j = low; j < high; j++) {
        if (arr[j] > pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return i + 1;
}
void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}
int main() {
    int n;
    printf("Enter number of sellers: ");
    scanf("%d", &n);
    int sales[100000];
    printf("Enter %d sales values: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &sales[i]);
    }
    quickSort(sales, 0, n - 1);
    printf("Sorted sales (descending): ");
    for (int i = 0; i < n; i++) {
        printf("%d ", sales[i]);
    }
    printf("\n");
    int sum = sales[0] + sales[1] + sales[2];
    int average = sum / 3;
    printf("Top 3: %d %d %d\n", sales[0], sales[1], sales[2]);
    printf("Average: %d\n", average);
    return 0;
}
```

Output

Clear

```
Enter number of sellers: 5
Enter 5 sales values: 42 36 98 76 45
Sorted sales (descending): 98 76 45 42 36
Top 3: 98 76 45
Average: 73


=== Code Execution Successful ===
```